

Partial Differential Equations 3 (Numerical)

Lecture 5 - Academic Year 2025/2026

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Aims and Objectives

1. Introduction to parabolic PDEs.
2. To discuss explicit time marching algorithm.
3. To explain von Neumann Stability Analysis.

Parabolic PDEs

We will now focus our attention to parabolic PDEs. Unlike the elliptic PDEs that we have described so far, which mainly concern with steady state solutions, many parabolic PDEs involve time evolution. A transient problem is one in which the solution changes in time. This class of problems is also often called time dependent, unsteady, non-equilibrium or non-steady problems. Different numerical approaches are needed from the ones we have discussed so far.

Examples of parabolic PDEs include:

- Heat conduction equation, $\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$.
- Time-dependent Schrödinger equation, $i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi$.
- If you are interested, the Black-Scholes equation in mathematical finance is also a parabolic PDE.

You may remember that in Lecture 1 one of the exercises is to check what type of PDE is the heat conduction equation. Actually, the PDE form above, with first derivative in time and second order derivative(s) in space, is extremely common for all sorts of diffusion type phenomena, making parabolic PDEs very important in science and engineering applications.

Explicit Time Marching: 1D Case

For simplicity and concreteness, let us consider a 1D heat conduction equation,

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2}. \quad (1)$$

As in previous lectures, we can use a central difference scheme to approximate the spatial derivative. The new element here is the time derivative. The simplest approach is to use explicit forward difference, and hence we can write the discretised 1D heat conduction equation as

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = \kappa \frac{T_{j+1}^n - 2T_j^n + T_{j-1}^n}{\Delta x^2}, \quad (2)$$

where we have used the superscript n to indicate the time step and subscript j to indicate the spatial grid. Fig. 1 illustrates the computational grid in time and space.

You may wonder at this point why we do not use a second order accurate central difference scheme for the time such that $\frac{\partial T}{\partial t} = \frac{T_j^{n+1} - T_j^{n-1}}{2\Delta t}$. We will come back to this later in the lecture

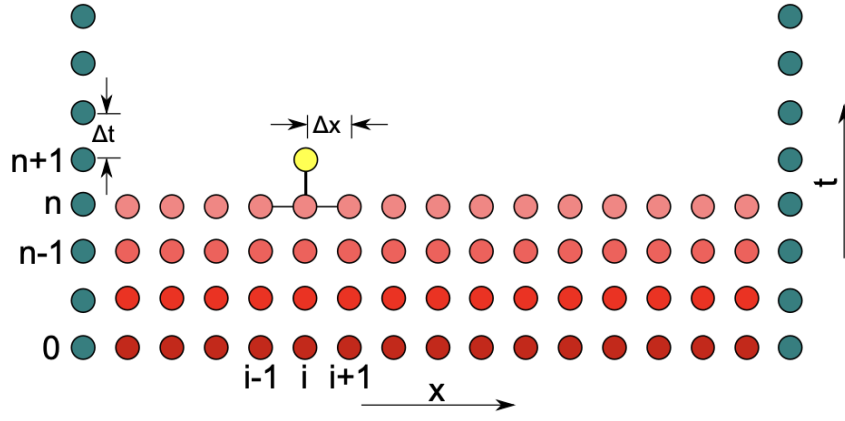


Figure 1: Example of a grid that includes time and 1D spatial discretisation.

after we discuss von Neumann stability analysis. As an exercise, you will show this leads to an unconditionally unstable scheme; thus, it is not useful.

Rearranging Eq. (2), we can find the FTCS (forward time-centered space) scheme:

$$T_j^{n+1} = T_j^n + \frac{\kappa \Delta t}{\Delta x^2} (T_{j+1}^n - 2T_j^n + T_{j-1}^n). \quad (3)$$

This scheme is formally first order accurate in time and second order accurate in space.

Exercise 1: Write the pseudo-code for a 1D FTCS scheme.

Explicit Time Marching: Higher Dimensions

The extension of the FTCS/explicit time marching scheme for 2D and 3D is straightforward. For instance, in the 2D case, the key algorithm becomes

$$T_{j,l}^{n+1} = T_{j,l}^n + \frac{\kappa \Delta t}{\Delta x^2} (T_{j+1,l}^n - 2T_{j,l}^n + T_{j-1,l}^n) + \frac{\kappa \Delta t}{\Delta y^2} (T_{j,l+1}^n - 2T_{j,l}^n + T_{j,l-1}^n). \quad (4)$$

Note that we avoid using i and k because we will need these letters - as customary - to describe the imaginary number and the wave number (see von Neumann stability analysis).

Example Problem

As an example, we consider a 1D heat conduction problem subject to the following boundary conditions:

$$\begin{aligned} T(0, t) &= 0 & \text{for } t \geq 0 \\ \frac{\partial T}{\partial x}(l, t) &= 0 & \text{for } t \geq 0 \\ T(x, 0) &= T_0 \sin \frac{3\pi x}{2l} & \text{for } 0 \leq x \leq l \end{aligned}$$

which has an analytical solution given by $T(x, t) = T_0 \exp\left(-\frac{9\kappa\pi^2 t}{4l^2}\right) \sin \frac{3\pi x}{2l}$. As before we have similar boundary conditions for the spatial coordinates. But now we need an additional boundary condition because we have a first derivative in time as part of our parabolic PDE.

Fig. 2 shows results from our FTCS method where we have used $\kappa = 1$, and two different grids: (i) $\Delta x = 0.08$ and $\Delta t = 5 \times 10^{-5}$; and (ii) $\Delta x = 0.01$ and $\Delta t = 5 \times 10^{-5}$. Clearly, the solution gets better with smaller mesh size and we can carry out a grid refinement analysis as before to determine mesh independent solution. At this point, I recommend you to play around with the interactive examples (see weblink below) where you can vary Δx for a given Δt , and vary Δt for a given Δx . Does the simulation break down at some point?

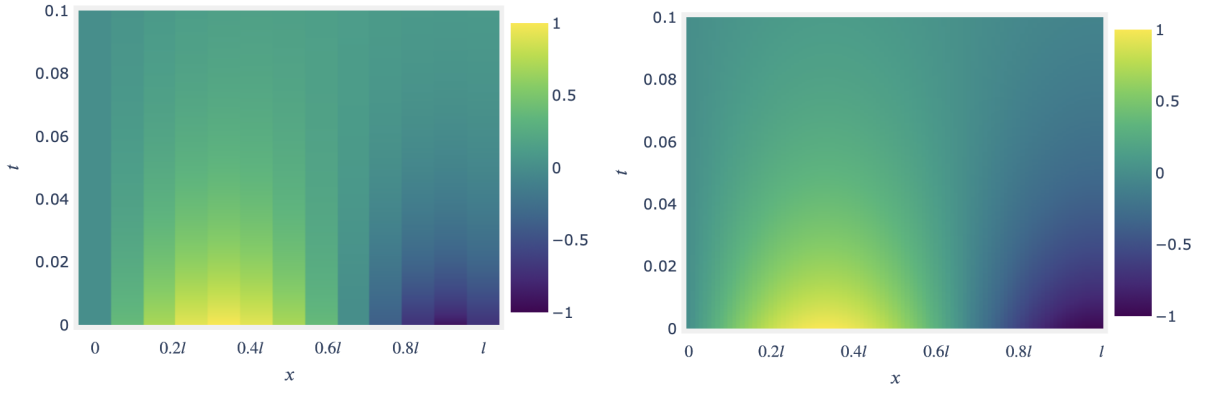


Figure 2: Example solutions using the FTCS method: (Left) $\Delta x = 0.08$ and $\Delta t = 5 \times 10^{-5}$; (Right) $\Delta x = 0.01$ and $\Delta t = 5 \times 10^{-5}$. Remember that time is plotted on the vertical axis, while space is plotted on the horizontal axis.

von Neumann Stability Analysis

You should have found that you cannot freely choose Δx and Δt in the FTCS scheme. While you can make Δt smaller and smaller for the same Δx , if you make Δx continuously smaller for the same Δt your calculation breaks down. Is this intuitive? How can we explain this?

The analysis tool to understand this is ascribed to John von Neumann (you may have seen his name in many areas of physics, mathematics, engineering and computer science). To do this, let us describe an error function,

$$\epsilon_j^n = (T_j^n)_{\text{num}} - (T_j^n)_{\text{discrete}}, \quad (5)$$

where ‘num’ describes the solution we would have got in our numerical scheme and ‘discrete’ describes the true solution of the equation in discrete space. Since both the ‘num’ and ‘discrete’ solutions satisfy the same discrete equation, the evolution of the error term also follows the same form as the original PDE scheme

$$\epsilon_j^{n+1} = \epsilon_j^n + r_x(\epsilon_{j+1}^n - 2\epsilon_j^n + \epsilon_{j-1}^n). \quad (6)$$

As a shorthand, I have introduced $r_x = \frac{\kappa \Delta t}{\Delta x^2}$. Now the idea of the Von Neumann stability analysis is actually quite simple. It seeks to investigate whether the error term keeps growing or not. If it does, then the proposed is unstable and hence should not be used.

For simplicity, we will do the analysis assuming we have a periodic boundary condition on the computational domain, as the result in the ‘bulk’ grid nodes does not change if we have Dirichlet or Neumann boundary conditions. By doing this, we can write things down in terms of Fourier series:

$$\epsilon(x, t) = \sum_{m=-M}^M E_m(t) e^{ik_m x}, \quad (7)$$

where the wavenumber $k_m = \pi m/L$ with $m = -M, \dots, -1, 0, 1, \dots, M$, $M = L/\Delta x$, and L is the interval of the computational domain. Substituting Eq. (7) to Eq. (6) for each k -mode, we get

$$\begin{aligned} E_m(t + \Delta t) e^{ik_m x} &= E_m(t) e^{ik_m x} + r_x (E_m(t) e^{ik_m(x+\Delta x)} - 2E_m(t) e^{ik_m x} + E_m(t) e^{ik_m(x-\Delta x)}) \\ g = \frac{E_m(t + \Delta t)}{E_m(t)} &= 1 + r_x (e^{ik_m \Delta x} + e^{-ik_m \Delta x} - 2) \\ g = \frac{E_m(t + \Delta t)}{E_m(t)} &= 1 - 4r_x \sin^2(k_m \Delta x / 2) \end{aligned}$$

The necessary and sufficient condition for the error to remain bounded is that $|g| = \left| \frac{E_m(t+\Delta t)}{E_m(t)} \right| = |1 - 4r_x \sin^2(k_m \Delta x / 2)| \leq 1$. For the condition to hold for all mode m , the maximum value of the $\sin^2$ term is one, and hence we get

$$r_x = \frac{\kappa \Delta t}{\Delta x^2} \leq \frac{1}{2}. \quad (8)$$

In numerical method analysis, the relation between Δx and Δt is often known as the Courant-Fredrichs-Lewey (CFL) condition. We can write

$$\Delta t \leq \frac{\Delta x^2}{2\kappa}, \quad (9)$$

or more generally

$$\Delta t = \nu \frac{\Delta x^2}{2\kappa}, \quad (10)$$

with $0 < \nu \leq 1$ is the Courant number. For safety we would normally choose $\nu = 0.95$ or smaller.

Be Careful: It is important to note that stable does not mean accurate. For example, we can make both Δx and Δt to be large. We can have a stable solution that is inaccurate.

Exercise 2: Show that taking a central difference scheme for the time derivative leads to an unconditionally unstable scheme.

Exercise 3: What is the von Neumann stability condition for a 2D FTCS scheme, see Eq. (4)?

Additional Materials

- Interactive tool: <https://kusumaatmaja.github.io/PDE3/examples>

Version Control:

- version 1.0 on 20 December 2025